Recycled E-Waste Miniature Climate Chamber

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Abstract

This project solves all 4 grand challenges by being 100% recycled, reducing electronic waste in the environment, recycling unused parts, and expanding the technological field for students in school laboratories. The solution that was chosen to solve all these grand challenges is the climate chamber. The climate chamber is a device that controls and measures temperature, humidity, and moisture inside in order to keep a specific environment for testing purposes. The prototype operates through two main functions, each divided into two linked actions. The first function controls temperature by generating heat and continuously sensing it. The second function monitors humidity by detecting moisture levels without active control. All sensor data is converted into electrical signals, allowing the display to present accurate temperature and humidity readings clearly and reliably. The main chamber (30x23cm) is made from aluminum and covered with wool insulation, and a computer fan for distributing the heat. The W1209 module is used to measure and control the temperature inside the chamber by measuring the temperature, and when it reaches the goal, the module switches on the fan. The sensor and module power supply is a 5.5V adapter, which means it is suitable for them and does not have an excess voltage. All the electronics are protected with acrylic sheets to prevent heat damage. The W1209 module and sensor system perform consistently across multiple test cycles. The error of the chamber reading temperature and humidity is less than 5% that means the chamber readings are accurate. This project is applicable in Biology and Chemistry laboratories.

Introduction

In the last century, humans faced the electronic revolution, which still occurs in our time. This has led to an escalating crisis in electronic waste. By 2021, the total amount of electronic waste reached 57.4 million tons, and it is predicted that the growth rate will be at least 2 MTs per year. Moreover, the experts estimate that there will be a total of 347 metric tons of non-recycled electronic waste by the end of the current year, which led the United Nations to come up with the phrase "E-waste tsunami." (**that shown in figure one**) The oil in the region surrounded by the electronics waste could get contaminated because of the large levels of lead and mercury that came from those materials, and it can be separated in the water and air, which disturbs the environment (according to the Soil contamination Inspection). Only 17.4% of the e-waste is collected and recycled, and the percentage is small and less than the quarter because in the developing countries they have weak management and a lack of the required technology. This pushes the countries to develop their technological infrastructure. From the previous, the main problem is defined as "recycling the electronic waste for economic and environmental purposes."

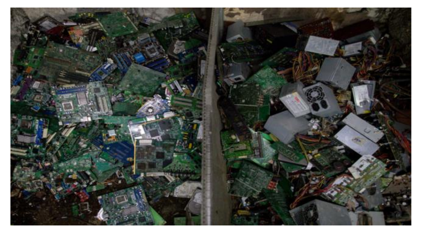


Figure 1: "E-waste tsunami."

Efforts were made to find a solution for this problem, either to end it or increase the amount of e-waste recycled. The most common solution among them all is disposing of them in the proper places, as it doesn't pollute the environment, and, if applied correctly, would make a great difference. However, most people in a third-world country wouldn't have enough awareness to let them do this, a rather easy job, and would end up throwing them in the street. Moreover, this solution could be a bit expensive for developing countries, as those bins must be spread everywhere to encourage people to get rid of waste in them. Furthermore, a lot of machines would be needed to recycle this waste, increasing the cost progressively, in addition to the initial cost of installing the bins.

The selected design requirement must have some properties ensuring efficiency and reliability. First of all, the solution must be eco-friendly, as this is the whole point of the project. Moreover, the error percentage mustn't exceed 5% ensuring accuracy. Lastly, the prototype needs to have a friendly user interface, as it should be easy to use by students in school labs.

The most suitable solution for this problem is the recycling of e-waste materials in old devices and utilizing them in building new devices, specifically ones used in school labs, as it would enhance the scientific and technological environment, in addition to solving the problem of e-waste. Moreover, taking parts from broken old devices and using them to make new ones would cut down the cost of building it from the very start. Also, it would save the environment, as some of those abandoned devices have valuable materials in them, meaning that each material extracted from an old device is another that would still be available in reserves.

All of this led to the selection of a climate chamber as it is the most suitable solution to meet the solution and design requirements. A climate chamber is a complex system that requires sensors, control logic, and a heating system. This device is widely used in schools, labs, and even hospitals due to its ability to simulate certain climates, enhancing chemical, physical, and biological experiments and control specific conditions (temperature and humidity) enhancing the experimentation process by testing the behavior of the microorganisms in this environment, and sometimes to prepare the medium conditions required for the reproduction of microorganisms. Although this device is easy to assemble, as it consists of a definitive subset of components, it's quite rare and limited in Egypt. This prototype would improve the technological environment in Egypt as it would be more available, benefiting more people and enhancing their experience in various experiments.

Materials & Methods

Container Construction:
The main part of the climate chamber is a recycled aluminium cheese container. Its dimensions are about 30 × 23 cm. The electronics are protected by an acrylic sheet which has been placed in between the heated area and the electronics. A heating coil from a thermostatic heater is affixed to the outer wall of the container and a recycled computer fan is set up to help mix warm air around the chamber thus avoiding any temperature difference.

Digital Components:
The hydrothermal sensor is tasked with monitoring the temperature and the humidity level inside the chamber. It is integrated into a system that consists of the W1209 thermostat module, which controls the heating coil. The first move is dialling in the desired temperature on the thermostat. The W1209 module will then trigger the heating coil when the sensor reports that the temperature is less than the defined level. The module will then turn off the heating coil once the temperature in the chamber matches the setpoint. The fan, thermostat, and sensor receive their power supply. The acrylic sheet serves not only as a separator but also as a protective shield for the electronic components during the operation of the system.

Material:
The heating system heats by using a coil that had previously been a part of a broken heater and thus recycled..

Name	Usage	Price
Aluminum Container	Housing	10 L.E (Recycled)
Computer Fan	Distribution of heat	50 L.E (Recycled)
Relay Module	Electronic switch	
W1209	Measures and controls the temperature of the device to keep the system at a set temperature	160 L.E (Recycled)
Hydrothermal Module	to measure the temperature and the humidity	180 L.E (Recycled)
Electrical Wire	To flow the electricity in the device	40 L.E (Recycled)
Acrylic Sheet	To isolate the electronics inside the device	150 L.E
Heating Coil	To flow the heat through the device	20 L.E (Recycled)
Power Plug	Supply power	10 L.E (Recycled)
Adapter 12v	Adapt power	60 L.E (Recycled)
Insulator	Insulate heat	

Test Plan:
First of all, the correctness of the temperature and humidity measurements was confirmed by comparing the hydrothermal sensor readings with those from a distinct reference device and checking the accuracy of our measurements in this way.
Secondly, the thermostat (W1209) test was initiated. After the temperature was selected, it was verified whether the heating coil could heat the chamber to that precise level.
• The automatic shut-off feature was tested as well in order to make sure that the heater did stop when the set temperature was reached.
• Subsequently, the circuit was monitored for voltage and current as a proof that the system was operating within safe limits and not drawing excessive power.
• To ensure the reliability of results, each test was performed three times.

Results

As the test plan informs, first we will set the thermostat(W1209) on a certain temperature, then it heats the heating coils to this specific temperature, the first test is to check if the heating coils temperature is elevated to the right value using a thermometer, then we measure this value using our temperature sensor and measure the relative error between them. The second test is measuring humidity in the air and measuring the relative error in comparison with the humidity measured by a trusted device. As shown in the opposing table

Positive results:
The system maintains set temperature and humidity within acceptable ranges for educational lab use like biology lab. Demonstrates that functional lab equipment can be built entirely from e-waste, supporting sustainability goals. The device is portable, easy to use, and suitable for students experiments in biology and chemistry labs.
There is no electrical hazards detected during operation; insulation prevents overheating of external surface. The W1209 thermostat module and sensor system perform consistently across multiple test cycles. The error of chamber reading temperature and humidity is less than 5% that means the chamber readings is accurate.

Negative results:
First we found the prototype is working but the temperature isn't perfectly uniform (slight hot spots may appear), and the heating speed is slower compared with industrial chambers.
The humidity can't be controlled without a dedicated dehumidifier, reducing humidity is very difficult; condensation could form inside. Using recycled materials affects on sensors accuracy over time, requiring periodic recalibration. Other recycled components also must be affected like the heating coil may have reduced lifespan compared to new parts.

Temperature set	Chamber reading (sensor)	Reference reading	Error percentage
30	30.90	30.10	3.00%
32	32.89	32.10	2.80%
34	35.05	34.20	3.10%
36	34.80	36.10	3.10%
38	38.98	38.19	2.60%
40	41.16	40.30	2.90%
42	43.21	42.18	2.90%
44	45.10	44.14	2.70%
46	47.24	46.2	2.70%
48	49.29	49.13	2.70%
50	51.55	50.2	3.10%

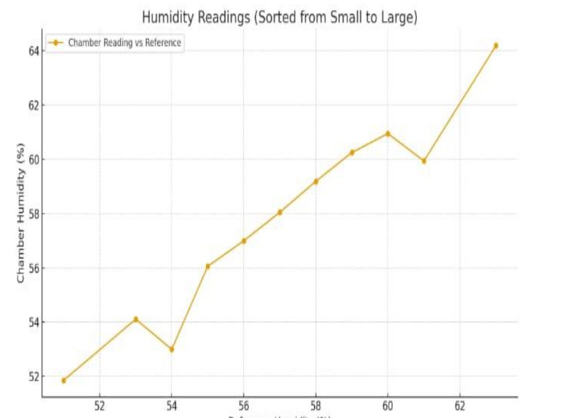
Reference reading (humidity)	Chamber reading (sensor)	Error percentage
53%	54.11%	2.10%
56%	57.00%	1.80%
58%	59.19%	2.06%
55%	56.06%	1.93%
60%	60.95%	1.59%
61%	59.94%	1.73%
54%	53.00%	1.84%
59%	60.25%	2.12%
63%	64.19%	1.89%
51%	51.86%	1.69%
57%	58.05%	1.85%

Analysis

According to the test results, the prototype and the performance appear through the design itself, where the project achieves a positive impact on both the grand challenge and achieves all the design requirements. Egypt produces an estimated amount of 100 million tons of trash annually. The composition of municipal solid waste is approximately 56% organic, 13% plastics, 10% paper and cardboard, 4% glass, and 15% other materials. Only 3.2 million tons are recycled, and this makes harmful effect on the climate for Egypt and the world.
To face this problem, the solution is a climate chamber, which is made almost entirely from recycled materials and is easy to get. The device measures the temperature and humidity and controls the temperature by using the thermostat. It works by plugging the prototype in the electricity, the electricity flows through the wires it reaches the heating coil and it takes the reading of the amount of heat by using the thermostat and by using PH.2.03 we determined the amount of voltage of the circuit by ohms law which was

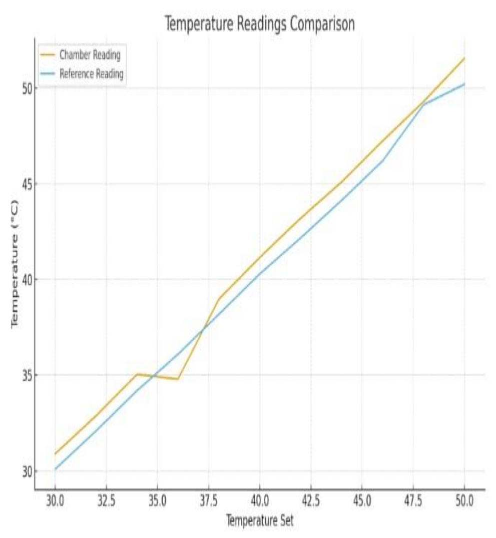
$$V = IR$$

there is a dimmer 2000 watt to decrease the amount of voltage from 220 volt to the heating coils to avoid using a lot of energy to just control the temperature from 30 degrees to 40 degrees. And the adaptor in the prototype is 5.5 volts and 2 amperes, which is used to convert the 220 volts to be able to connect to the w1209 module and the humidity sensor. The module needs 12 volts; therefore, there are step up convertor to increase the voltage from 5.5 to 12 volts. The humidity sensor needs 1.5 volts; therefore, there are 15 kilo ohm resistance connected to decrease the voltage from 5.5 to 1.5 volts.
The test of the project was made. An old aluminum container was used as the main chamber because it spreads heat well. To help distribute the heat more effectively, I installed a 12-volt fan connected with the module w1209 to switch on and off the fan when the temperature sensor in the module reads the goal temperature, so the temperature wouldn't gather in one spot and would spread Fastly. Also, humidity sensors were added (old hygrothermometer), with the readings shown on an LCD screen. This sensor is important because it measures the humidity of the surrounding air, which can be shown by the humidity sensor, to make sure the environment stays stable.
For controlling the temperature module w1209 was connected to allows set the exact range it needs. Heating coils were put inside the aluminum container to act as the main heat source, while the module automatically turns the fan on and off to achieve the chosen temperature. All components were connected using proper wiring to ensure that everything receives stable power and works safely.
The idea behind the device is to create a controlled space where heat is produced, monitored, and adjusted continuously. The fan spreads the heat evenly, the heating coils supply the heat all over the chamber, and the module keeps the system under control. At the same time, the sensor provides constant feedback about humidity that affects the device, making sure that these readings are accurate and that the system runs as the real one we actually need
After designing and assembling the thermal control chamber using a repurposed aluminium container, a 12V fan for heat distribution, a humidity sensor with a display, heating coils, a module w1209 for precise control, a dimmer to decrease the voltage, and an adaptor to be the power supply of the module and the sensor, the system was tested for accuracy. The readings from the internal sensor were compared with those from a
stability.
After this test was made, there were results, as shown in graph 1.
The humidity graph shows that the chamber sensor closely follows the reference device. Both exhibit a similar rising trend over time, confirming that the internal environment always changes. Minor deviations between two lines are expected because the dynamic nature of humidity in Egypt changes from one place to another, especially with active air circulation from the fan. The sensor reliably tracks the reference trend, indicating stable performance



Graph 1: The relation between the measured reading and the reference reading.

As shown in graph 2:Temperature Readings
The temperature graph between the chamber sensor and the reference. Both lines rise together when the heating coils are active, reflecting effective thermal control. Here, (MA.2.01) (Polynomial Functions) was particularly useful in analyzing the temperature curve. The smooth, gradual increase can be modeled as a polynomial function, helping to understand the system's thermal behavior and validate the sensor's response against the reference data.



Graph 2: The relation between the measured value of temperature and the reference value.

To evaluate accuracy, the percent error was calculated for both humidity and temperature using this formula:

$$\text{Percent error} = \left|\frac{\text{average Reference value} - \textit{average}\ \text{Measured value}}{\text{avergae Reference value}}\right| \times 100$$

For the humidity average of the measured value was **57.69%** compared to the reference 57**%**, yielding:

$$\left|\frac{57 - 57.69}{57}\right| \times 100 \approx 1.21\%$$

•For the temperature average of the measured value measured value was 40.9°C versus the reference 40°C, yielding:

$$\left|\frac{40 - 40.9}{40}\right| \times 100 \approx 2.25\%$$

Both errors are less than 5%, confirming the system's high accuracy and reliability.

Conclusion

Based on the resulting analyses, the prototype successfully met the design requirements and was able to measure temperature, humidity, and temperature changes. Furthermore, the results:

- showed that the prototype has the potential to solve and influence our problem, as measurements demonstrated an accuracy of 97.75% for temperature and 98.79% for certain types of samples.
- The entire prototype is composed of recycled electronic components salvaged from unused devices, such as power supplies and heaters.
- The prototype is suitable for use in chemistry and biology labs.
- Our prototype can solve the problem of discarding and burning electronics, which leads to death, various diseases, and pollution.
- This project has the potential to be a sustainable solution to major challenges, such as reducing pollutants that contaminate air, water, and soil. It also allows for the recycling of unused components instead of throwing them in landfills, improving the work environment and technology at a low cost and with a high impact.

Recommendations

- Adding suitable housing and insulation is necessary because the housing is small and cannot accommodate all the samples needed, and it also conducts heat. Adding more insulating housing will provide greater safety and better results. Hollow polycarbonate can also be used; it is transparent, allowing you to see the contents inside and make modifications.
- Using Arduino UNO/PCP boards allows for more and easier applications, such as designing programs that work via Bluetooth or Wi-Fi, which saves time compared to collecting data on paper. This didn't fit into the prototype; the reason these components weren't available in any electronic waste.
- Adding a complete cooling system will allow for more uses in school laboratories in the future, and new samples can be placed inside, thus increasing the device's effectiveness. Due to design requirements and the complexities of assembling any device, it may need to be disassembled.
- Using high-quality sensors will have a positive impact; when you test the project, it will give high accuracy, which will help with the calculations in the project.

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